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Computational Modeling of Dissimilar Metal CO₂ Laser Welding: Applied to Copper and 304 Stainless Steel

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Abstract

This research work analyses the thermal, metallurgical and physical stages of AISI 304 Stainless Steel-Copper dissimilar couple during laser welding in keyhole mode numerically and then validated experimentally. 10.6 μm wavelength CO₂ laser welding machine is used for conducting this experimental analysis and the process is simulated by finite volume method. For making the dissimilar couple, a volumetric Gaussian heat source is applied symmetrically on the both metal domain and it is getting an asymmetric molten pool shape. It is observed that the computational model have good concurrence with the experimental results, after the calibration between simulation and experimental results for the same parameter set.

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1. Introduction

Laser welding is an advanced non-conventional welding process to join the metals/non-metals and have wide range of industrial applications. Laser welding is one of the most efficient energy beam welding process for producing high energy density impingement on a concentrated weld surface. These in-turn results in thin welded area, thin heat affected zone and thus minimum distortion. The main advantage of continuous wave laser welding is to reach deep penetration even with high weld beam travelling speed. For modelling the welding, Rosenthal proposed the first quasi-steady state mathematical model for moving heat source [1]. After Rosenthals proposal, different studies had been carried out in [2–4] to explore the characteristics of heat transfer, stress distribution and transport phenomena of weld topology. Generally, welding simulation and analysis can be categorized by heat transfer based on the pure conduction model [5–7], heat transfer and transport phenomena of welding incorporated with convection (fluid flow) [8–13] and the consecutive coupled analysis of thermo-mechanical problem focussing on metallurgical and structural characteristic (pure conduction model with stress distribution) [4]. Conduction mode welding and key hole mode (penetration) welding are the one of the two fundamental category of laser welding, where pulsed [2–7] and continuous welding are the other types.

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In the continuous wave [8–13] welding process, the work piece is incident by a continuous wave laser beam of adequate intensity and moves at a constant velocity. The work piece absorbs a portion of the incident energy and generates a weld pool [8], which extends and solidifies as the laser beam moves away. Many experimental-numerical two dimensional and three dimensional models for the laser welding in all modes were presented to study the characteristics related to the transport phenomenon occurring in similar and dissimilar welds of different materials for butt weld geometry [10,12]. In [9], a detailed description about the laser beam welding of dissimilar metal (copper-nickel couple) was presented. The thermo-physical characterisation of weld pool and its effects are analysed numerically in their model. Recently, a coupled analysis of CFD and FEM model for predicting the weld pool geometry in laser welding with the help of FLUENT® software package was developed in [13].

The requirement of dissimilar metal joining dominates a major role in various emerging manufacturing applications. The welding of two different metal combinations is a challenging task due to the enormous dissimilarities in chemical and physical properties of the metals present. Hence, the main objective of this paper is to analyse the transport phenomena of Cu and AISI 304 Stainless Steel dissimilar couple. The complexness of the interactions concerned during this process and the succeeding behaviour of molten metal generated in a solid by incident laser radiation inhibit a completely perplexing simulation of laser welding. A number of assumptions are embraced in calculations of heat and mass transfer during laser beam welding to simplify this complex problem. The numerical based analysis gives an idea of the impact of operational parameters on the final weld joint and also helps to realize the heat transfer and flow profiles.

Nomenclature

p	Pressure
ρ	Density
μ	Viscosity
t	Time
ξ	Liquid fraction
δ	constant
λ	Thermal Conductivity
T	Temperature
H	Enthalpy
L	Latent Heat of Fusion
A_m	Mushy zone (solid-liquid transformation) parameter
c	Specific Heat
I	Input Laser power
η	Laser absorptivity of the material
σ	Surface tension
r	Laser beam radius
d	Thickness of the plate
U	Laser scanning speed

2. Numerical and Mathematical Formulation

A schematic of the laser welding setup for numerical and experimental is shown in Fig 1(a). For numerical analysis, the computational domain is considered as a three dimensional Cartesian co-ordinate system with fixed grid topology. The molten metal flow inside the weld pool was considered to be laminar in the early stages of the computations and later it was considered to be turbulent in nature. The general momentum conservation equation for the weld pool can be written as [13]:

$$\rho \frac{\partial u_j}{\partial t} + \rho \frac{\partial u_i u_j}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\mu \frac{\partial u_j}{\partial x_i} \right) - \frac{\partial p}{\partial x_j} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_j}{\partial x_j} \right) - A_m \left(\frac{(1-\xi)^2}{\xi^3 + \delta} \right) u_j - \rho U \frac{\partial u_j}{\partial x_j} + \rho g \beta (T - T_a) \quad (1)$$

where p, ρ, t, μ and ξ represent the pressure, density, time, viscosity and the liquid fraction of the domain respectively. In the above equation x_i, x_j and u_i, u_j are the distance and velocity component along the i and j directions respectively, δ is a constant (used to avoid division error), A_m is the Mushy zone (solid-liquid transformation) parameter ($A_m = 1.6 \times 10^4$) and U is the laser scanning velocity. The fourth term in the RHS of the above equation is used to handle the solidification around the solid-liquid transformation interface region of the weld pool. The enthalpy-porosity method is employed to solve the solidification phenomenon numerically. The fifth term signifies the relative motion between the laser welding source and work-piece. The sixth term stands for buoyancy. The temperature profile can be acquired by solving the following energy equation [9]:

$$\rho \frac{\partial H}{\partial t} + \rho \frac{\partial u_i H}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{\lambda}{c} \frac{\partial h}{\partial x_i} \right) - \rho U \frac{\partial h}{\partial x_i} - \rho U \frac{\partial \Delta H}{\partial x_i} + Q'' \quad (2)$$

where λ is the thermal conductivity of the material and H is total enthalpy. H is the summation of sensible enthalpy (h) and latent heat content (ΔH), which is utilised to follow the phase transformation.

$$H = h + \Delta H \quad (3)$$

$$h = h_a + \int_{T_a}^T C dT \quad (4)$$

$$\Delta H = \xi L \quad (5)$$

where T is the temperature, C is the specific heat and h_a, T_a are the reference enthalpy and reference temperature. L is the latent heat of fusion; the liquid fraction ξ in the mushy zone (solid-liquid transformation) assumed with linear approximation between the solidus and liquidus temperatures and is given as:

$$\xi = \begin{cases} 1, & \text{for } T > T_{Liq} \\ \frac{T - T_{sol}}{T_{Liq} - T_{sol}}, & \text{for } T_{sol} \leq T \leq T_{Liq} \\ 0, & \text{for } T < T_{sol} \end{cases} \quad (6)$$

where T_{sol} and T_{Liq} are the solidus and liquidus temperatures respectively. Liquid fraction value, ξ , of each cell among the computational domain was used to trace the liquid-solid interface (mushy zone) that is calculated in each iteration, based on an enthalpy balance. The phase transformation zone is modelled as a "pseudo" porous medium wherever the porosity changes from 1 to 0 as the material solidifies. Once the material has completely solidified in a

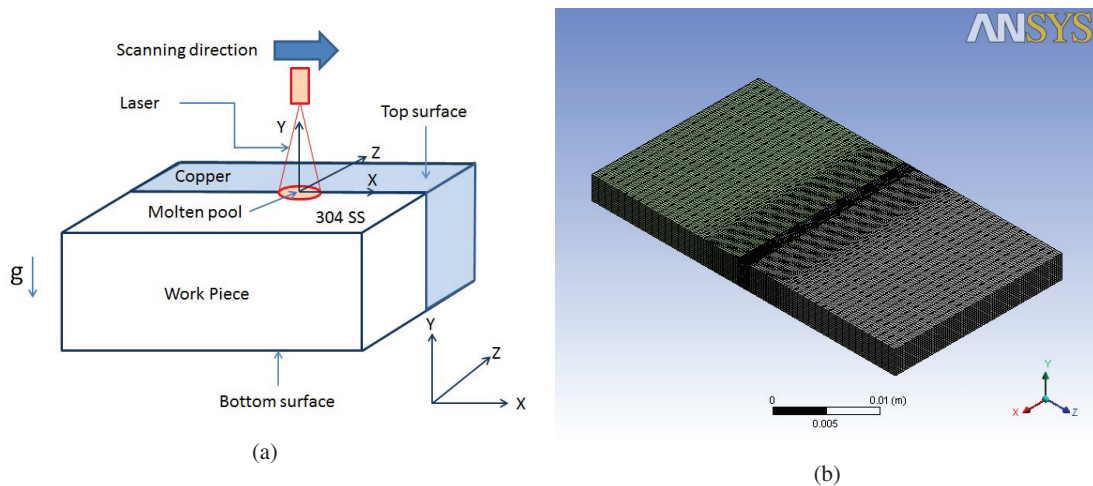


Fig. 1: (a) A schematic diagram of the laser welding setup (b) Mesh used for the analysis

cell, the porosity turn out to be zero and so the velocity converges to zero. When a laser beam is impinged on material surface, some percentage of incident beam energy is reflected. The absorbed amount of energy is transferred from the surface to interior by conduction heat transfer at the earliest stage of the process then the material surface start to melt and it will propagate into the work-piece by convection. A three dimensional Volumetric Gaussian heat source [13] was used to model the incident laser beam interaction with the material can be state as:

$$Q'' = \frac{3I\eta}{\pi dr^2} \exp\left(-\frac{3(x^2 + y^2)}{r^2}\right) \exp\left(\frac{-3z^2}{d^2}\right) \quad (7)$$

where I is the input laser beam power, η is the laser power absorptivity of the material, d is the thickness of the plate, r is the incident beam radius and x, y, z are the co-ordinates of the domain in X, Y and Z direction respectively. Heat loss as a result of radiation and convection is considered over the entire surface. The boundary condition at the free surface is expressed below:

$$\mu \frac{\partial u}{\partial z} = \xi \frac{\partial \sigma}{\partial T} \frac{\partial T}{\partial x} \quad (8)$$

$$\mu \frac{\partial v}{\partial z} = \xi \frac{\partial \sigma}{\partial T} \frac{\partial T}{\partial y} \quad (9)$$

$$\mu \frac{\partial w}{\partial z} = \xi \frac{\partial \sigma}{\partial T} \frac{\partial T}{\partial z} \quad (10)$$

where u, v and w are the velocities along the direction X, Y, and Z respectively, which are determined from the Marangoni convection effect. The boundary conditions in the vertical plane were defined as zero flux across the surface (i.e. at $y = 0$). For the other surfaces, the temperature was taken as the ambient temperature and the velocity was set to zero. The temperatures of the solid surfaces far away from the heat source were set to ambient temperature, T_a and at zero velocity. The 3-dimensional coupled continuity, momentum, energy and mass fraction equations along with the boundary conditions are solved numerically using a finite volume technique which is performed in the ANSYS 15 FLUENT[®] environment. The mathematical model used in this work was based on the Reynolds method of averaging with time dependant equations (RANS). Realizable $k - \xi$ turbulent model is implemented for calculating the viscous flow variables in the computational domain. The evaporation is not considered in this analysis and the weld pool is assumed to be in molten stage only. The general structure of the numerical solution lays on SIMPLER algorithm, modified appropriately to accommodate phase change process and mixing of dissimilar metals. Transient studies are carried out till some mixing patterns are obtained.

A grid system of variable spacing need be utilized with fine grid near the molten pool and coarse grid away from the molten pool as shown in Fig 1(b). The material was assumed to be homogeneous and isotropic with temperature dependent material properties adopted from Smithells Metals Reference Book [14]. The experiments and simulations are performed, with a speed of 4 mm/min and a constant laser power of 3000 W. In-line with the experimental parameter, the laser spot diameter of the beam over the sample surface was maintained at 180 μ m. A compiled User Defined Function (UDF) file was used to perform the volumetric laser heat source and to trace the weld pool surface profile, which is written in the C programming language associated with the ANSYS FLUENT[®] solver.

3. Results and Discussion

A portion of the laser power incident on the metal surface is absorbed by the metal and the heat in this way generated is transferred by thermal conduction and afterward by convection. The temperature and velocity contours of the considered model are presented to show the significance of transport phenomenon. Fig 2(a), Fig 2(b) and Fig 2(c) show the development of the weld pool as the laser heating happens after 1 ms, 1.4 ms and 10 ms. From the figure it is clear that as the welding procedure proceeds, the metal melts and the consequent fluid convection leads to a deformation on the melt pool free surface. The results illustrate that the starting periods of laser heating process, the AISI 304 Stainless steel getting a quick rise in temperature than copper. This is due to the fact that AISI 304 Stainless steel offering poor thermal diffusivity compared to pure copper. As a result, the location of maximum temperature

will be shifted towards the steel side and thus the steel melts initially and its poor thermal diffusivity compensated with its high melting point.

The cross sectional temperature contour view of the fully developed pool shape after 10 ms of welding process during the heating cycle is shown in Fig 3. The velocity contour on the material top surface for a fully developed weld pool is shown in Fig 4. The figure shows the flow pattern with a maximum velocity almost 2.5 m/s, which is very higher than the laser scanning velocity. Among the forces that influence the molten pool dynamics pressure and surface tension are leading forces, as they initiate the flow within the molten pool. The temperature of the melt pool at the centre line is higher than at the edges and hence the surface tension is higher towards the edge lines. This effect causes the fluid particles to pull radially outward from the centre. Thus the melt pool advancement is more towards the radial direction compared to the melt pool development towards the downward direction. AISI 304 Stainless steel melts initially and is convected towards the copper side through the surface-tension driven Marangoni flow. Hence, the start of melting on the copper side is aided by the flow of metal from the steel area, besides direct heating by the laser. The transient formulation utilized as a part of the present work permits one to take after the development of the weld pool in this way. At later times moreover, the flow pattern is dominated by the solid Marangoni convection starting from the steel side.

The fully developed pool shown in Fig 2(c) and Fig 3 depicts an asymmetric shape. In light of symmetric heating and fluid convection, the isotherms twist and get to be non-circular. The continuous molten pool turns out to be fully developed after the beginning phase of advancement. Hence, in a reference frame moving with the laser (as used in our computational model), the pool reaches a quasi-steady state once it is fully developed. In the following phase of numerical simulations, the laser heat is turned off with the purpose of study the solidification phenomena for the period of the cooling stage. The size of the base metal is considerably large as compared to the molten pool and hence acts as a heat sink. Thus the cooling happens quickly and eventually the weld pool shrinks; leading to a completely solidified

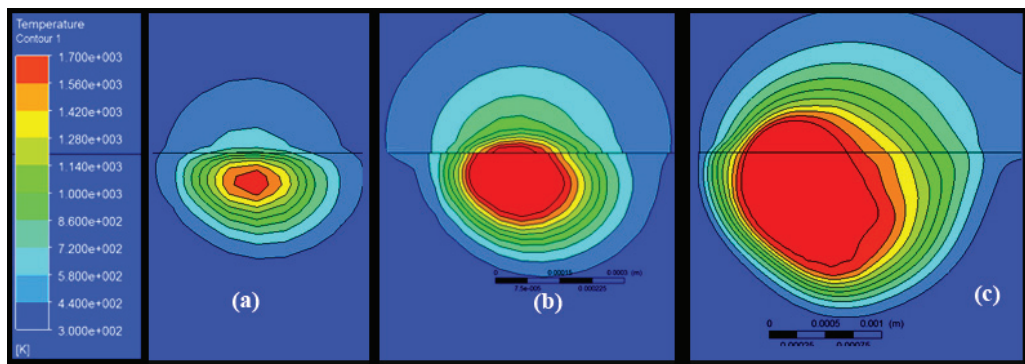


Fig. 2: Temperature contours of the top view of weld pool after (a) 1 ms, (b) 1.4 ms and (c) 10 ms

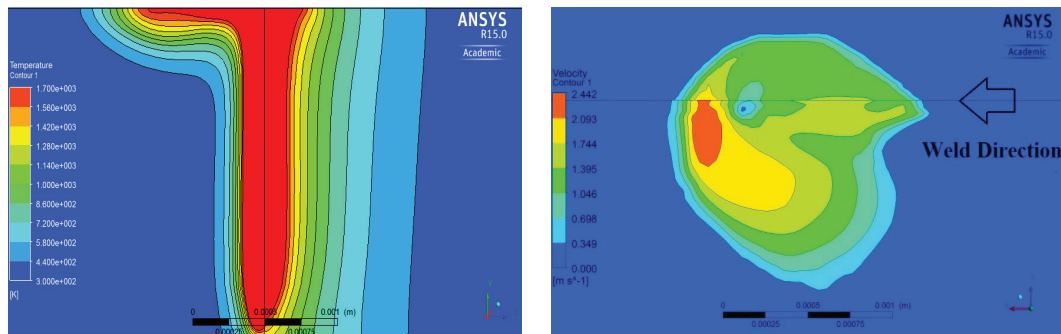


Fig. 3: Temperature profile of the cross-sectional view at fully developed weld pool after 10 ms

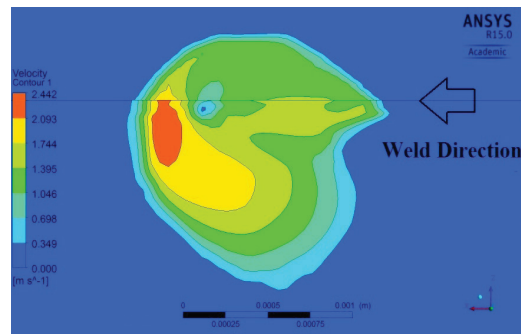


Fig. 4: Velocity contour of the top surface at fully developed weld pool after 10 ms

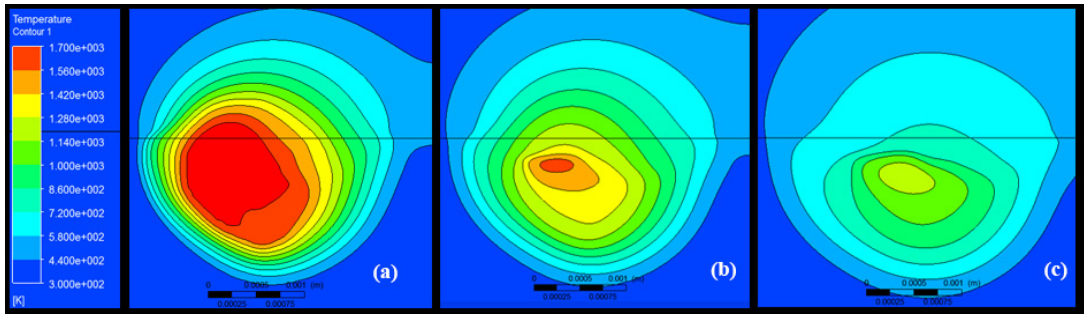


Fig. 5: Temperature contours in the top view during the cooling cycle: after (a) 0.8ms, (b) 5.8ms and (c) 11ms

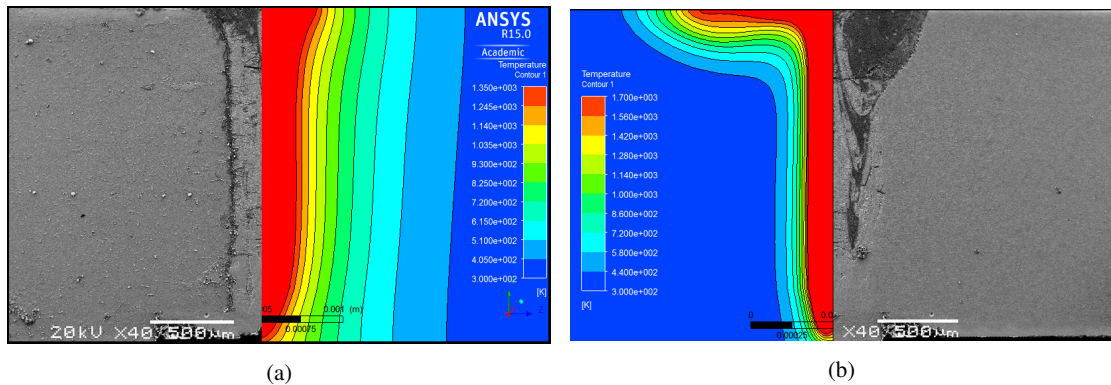


Fig. 6: Simulation V/S experimental results (a) copper and (b) steel

weld. It might be noticed that the liquid flow in the weld pool is fundamentally determined by thermal buoyancy effects subsequent to the high temperature gradients on the weld pool surface vanished when the laser power is turned off. The free surface temperature contours during the cooling process after a time interval of 0.8 ms, 5.8 ms and 11 ms are shown in Fig 5(a), Fig 5(b) and Fig 5(c) respectively.

The experimentally attained molten pool shapes of copper and 304 stainless steel shown in Fig 6(a) and Fig 6(b) respectively, confirm the numerical calculation furthermore, typically specify a deep penetration welding i.e. immediate transport of heat inside the material volume as the beam impinges on the 2 mm thick metal plates for laser power 3000 W, welding speed 4 mm/min. and beam angle 90° . All the figured molten pool shapes are similar to the experimentally measured weld pool shapes with a smaller percentage error standard of 5.982, which is well within the acceptable limit. After a calibration stage between simulation and experimental results for the same workplace, the ANSYS FLUENT® model has been great concurrence with the experimental results.

4. Conclusion

In this paper, an ANSYS Fluent® CFD model for laser welding of dissimilar metal has been developed for predicting and analysing the weld bead geometry. The considered dissimilar material welding model consists of copper - steel material using continuous wave CO_2 laser and the analysis has been studied numerically and is validated with experimental results. The considered model has to extract some key features of the process such as differential heating of the dissimilar metals, development of asymmetric weld pool, and mixing of molten metals. The acquired numerical results point up a good quantitative concurrence with the consequent experimental ones. This work lays a solid foundation for future studies of some of the complex issues in dissimilar joints.

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